

Literature Review — SENTINEL XDR: An LLM-Augmented Intrusion Detection System

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1. Introduction to the Problem Domain

Intrusion Detection Systems (IDS) monitor network or host activity for malicious behaviour and policy violations. Classical IDS fall into two families: **signature-based** detection, which matches traffic against a database of known attack patterns, and **anomaly-based** detection, which models "normal" behaviour and flags deviations [1], [10]. Signature systems are precise for known threats but blind to novel attacks; anomaly systems generalise to new threats but suffer from high false-positive rates [4]. More recently, **machine-learning IDS (ML-IDS)** and, in the last two years, **Large-Language-Model (LLM) augmented** security tooling have emerged to bridge the gap between detection coverage and analyst interpretability [7], [8]. This review surveys eight representative works across these categories, identifies their limitations, and positions the SENTINEL XDR project against them.

2. Review of Related Work

2.1 Signature-Based IDS

[1] **Roesch (1999) — Snort**. Snort introduced a lightweight, rule-based network IDS that inspects packets against a community-maintained signature set. It remains the de-facto baseline for signature IDS. *Strength*: high precision and low false positives for known threats. *Limitation*: purely reactive — it cannot detect zero-day or obfuscated attacks, and rules require constant manual curation.

[2] **OISF (2010-) — Suricata**. Suricata extended the Snort model with multi-threading, protocol/flow awareness, and TLS metadata extraction, improving throughput on modern multi-core hardware. *Limitation*: it inherits the signature paradigm's inability to reason about *multi-stage* campaigns and still produces low-context alerts that human analysts must manually correlate.

2.2 Machine-Learning & Anomaly-Based IDS

[3] **Sharafaldin et al. (2018) — CICIDS2017**. This work produced a widely used labelled dataset and benchmarked classifiers (Random Forest, KNN, etc.) for flow-based intrusion detection. *Contribution*: a realistic, modern benchmark. *Limitation*: offline, flow-level classification; the models output a class label with no explanation, MITRE mapping, or remediation guidance.

[4] **Buczak & Guven (2016) — ML/Data-Mining for IDS (survey)**. A comprehensive IEEE survey of ML methods (SVM, decision trees, clustering, neural nets) for cyber intrusion detection. *Key finding (and gap)*: model accuracy is

dataset-dependent and the *interpretability* of ML-IDS output is consistently cited as an unsolved operational problem.

[10] Mirsky et al. (2018) — **Kitsune**. An unsupervised, online anomaly IDS using an ensemble of autoencoders for plug-and-play deployment without labelled data. *Strength*: online, light-weight, no signatures. *Limitation*: anomaly scores are not human-readable and give no attack attribution or recommended response.

2.3 Knowledge Frameworks & Retrieval

[5] Strom et al. (2018) — **MITRE ATT&CK**. ATT&CK is a curated knowledge base of adversary tactics and techniques (TTPs). It is the lingua franca for describing attacker behaviour but is *not itself a detector* — it must be operationalised by a detection system.

[6] Lewis et al. (2020) — **Retrieval-Augmented Generation (RAG)**. RAG couples a parametric LLM with a non-parametric retriever over an external corpus, improving factual grounding for knowledge-intensive tasks. *Relevance*: RAG is the mechanism by which an LLM can cite up-to-date ATT&CK knowledge rather than hallucinating it.

2.4 LLM-Augmented Security

[7] Gupta et al. (2023) — **"From ChatGPT to ThreatGPT" (IEEE Access)**. Surveys both the offensive and defensive impact of generative AI in cybersecurity, arguing LLMs can accelerate triage, summarisation, and analyst assistance, while warning of prompt-injection and hallucination risks. *Gap*: it is a survey — it does not deliver an end-to-end LLM-driven detection pipeline.

[8] Ferrag et al. (2024) — **Generative AI / LLMs for cyber threat detection**. Recent work explores fine-tuned and prompted LLMs for malware and intrusion analysis, reporting promising classification and explanation quality. *Limitation*: most studies evaluate the LLM in isolation, decoupled from a live packet-capture and correlation pipeline, and rarely address multi-provider resilience or evasion of the LLM itself.

3. Comparative Analysis

Work	Type	Detection	Explainable output	MITRE mapping	Multi-stage correlation	LLM/RAG	Live capture	Autonomous response
Snort [1]	Signature	Known only	✗	✗	✗	✗	✓	✗
Suricata [2]	Signature+flow	Known only	✗	partial	✗	✗	✓	✗
CICIDS2017 ML [3]	ML (offline)	Known classes	✗	✗	✗	✗	✗	✗
Kitsune [10]	Anomaly (online)	Novel	✗	✗	✗	✗	✓	✗
ML-IDS survey [4]	ML	Varies	✗ (noted gap)	✗	✗	✗	varies	✗
ThreatGPT [7]	LLM survey	—	✓	✓	✗	partial	✗	✗
GenAI-IDS [8]	LLM	Known/novel	✓	partial	✗	partial	✗	✗
SENTINEL XDR (ours)	Hybrid heuristic + LLM + RAG	Known + novel	✓ (NL + MITRE)	✓	✓ (MACE)	✓ (NVIDIA→Gemini→Groq + ChromaDB)	✓ (Scapy)	✓ (ADRS dry-run)

4. Identified Research Gaps & How SENTINEL Addresses Them

1. **Interpretability gap** [4], [10]: classical/ML IDS emit opaque scores or class labels. → SENTINEL pipes each flagged event through an LLM "SOC analyst" that returns a natural-language explanation, confidence, and a recommended action.
 2. **Knowledge-grounding gap** [7]: standalone LLMs hallucinate. → SENTINEL uses **RAG over a 25-technique MITRE ATT&CK corpus (ChromaDB)** so explanations cite real TTP IDs.
 3. **Multi-stage blindness** [1], [2], [3]: per-packet/per-flow detectors miss campaigns. → The **MACE** engine correlates atomic alerts into kill-chain attack chains.
 4. **Resilience gap** [8]: single-provider LLM tools fail on rate-limits/outages. → A **three-tier provider fallback** (NVIDIA NIM → Gemini → Groq) with exponential back-off.
 5. **Response gap**: most academic IDS stop at detection. → **ADRS** proposes (dry-run) containment policies, and **CHRONICLE** auto-generates executive incident narratives.
 6. **LLM self-defence**: LLM tooling introduces a new attack surface (prompt injection). → The **AEGIS** engine scans for adversarial/prompt-injection patterns.
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5. Conclusion of the Review

The literature shows a clear trajectory: signature IDS → ML/anomaly IDS → LLM-augmented security. Each step improved coverage but left **interpretability, grounding, multi-stage correlation, and response** only partially solved. SENTINEL XDR synthesises a **live heuristic detector**, a **RAG-grounded multi-provider LLM analyst**, **multi-stage correlation (MACE)**, and **autonomous (dry-run) response (ADRS/CHRONICLE)** into one pipeline — directly targeting the gaps catalogued above.

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